GIN TUANG

Evanston, IL | 918-606-6791 | tuanggin0@gmail.com

Mechanical Engineering Student | Mechanical Design · Analysis · Manufacturing · Automation

Mechanical engineering student who designs parts in CAD, backs decisions with FEA, hand calcs, and test data, and builds software that makes engineering work faster. Grew up in a manufacturing household and currently building CAD automation tools in production R&D.

EDUCATION

Northwestern University, Evanston, IL

Bachelor of Science in Mechanical Engineering | GPA: 3.4 — Expected June 2028

Relevant Coursework: Mechanical Design & Manufacturing (ME 240), Theory of Machines (ME 315), Manufacturing Processes (ME 340-1), Mechanics of Materials, Stress Analysis, Materials Science, Dynamic Systems, Probability & Statistics, Thermodynamics, Fluid Mechanics, Heat Transfer, Numerical Methods

EXPERIENCE

The AAON Group — Tulsa, OK

R&D Programming Intern — Full-Time → Part-Time (Remote) — June 2026 – Present

- Built the collision-detection and region-optimization logic for a Python CAD automation engine that auto-places components (fuses, control boards, VFDs, power distribution blocks) into 3D Inventor assemblies across the RNA–RND and RQ electrical panel lines.
- Built the pipeline from scratch to convert electrical DWG drawings into 3D CAD assemblies using the Inventor and Solid Edge APIs, replacing a manual layout step with a repeatable, rule-based process.
- Presented a 10-minute business case to engineering leadership projecting reductions in wire cost, material waste, and floor-level errors from automated layout.
- Worked in a production codebase with Git and Azure DevOps: submitted pull requests, completed merges, and reviewed teammates' code.

Barnes & Noble College (Northwestern University) — Evanston, IL

Student Associate — Nov 2025 – June 2026

- Handled inventory flow and POS/digital order processing during high-volume periods, keeping the store stocked and orders error-free.

PROJECTS

Bike Brake Caliper Design — Northwestern University (Team 19) — Spring 2026

FEA/Topology Lead

- Led FEA and topology optimization across 3 design iterations for a 3D-printed dual-caliper brake, cutting peak stress from 126 MPa to 34 MPa at 42.1 g.
- Root-caused a prototype failure by comparing physical test results against FEA predictions, then redesigned the geometry (thickness, fillets, hole placement) to meet ISO 4210 braking and durability requirements.

Northwestern Formula Racing (FSAE) — Aero Team Member — Aug 2025 – June 2026

- Designed rear wing inserts using topology optimization, FEA, and CFD, achieving a 33% mass reduction while holding FoS 2.0+ under peak aerodynamic loading.

Dyson Ring Structural & Thermal Analysis — Northwestern University (ME 362) — Spring 2026

- Ran structural FEA in ANSYS on a two-material concept under combined orbital, radiation-pressure, and thermal loads; found that real materials failed and quantified the material-property gap needed for a feasible design.

Two-Stage Gear Transmission Design — Northwestern University — Fall 2026

- Designing gear 4 and the output shaft (shaft 3) of a two-stage, three-shaft gear transmission on a three-person team, applying Shigley machine design methods.

SKILLS

- **CAD & Drawings:** Siemens NX, SolidWorks, Autodesk Inventor, Solid Edge, GD&T, Tolerance Analysis
- **Analysis:** FEA (ANSYS), Topology Optimization, CFD, Hand Calculations, Failure Analysis
- **Prototyping & Manufacturing:** 3D Printing, Physical Testing, DFM, Sheet Metal Forming, Tensile Testing
- **Programming:** Python, MATLAB, Excel, Inventor/Solid Edge APIs, Git, Azure DevOps
- **Languages:** English, Zopau (Oklahoma Seal of Biliteracy)

LEADERSHIP & SERVICE

- **Youth Camp & VBS Counselor, FGA Int'l:** Led 15+ youth sessions and community programs for 300+ participants.